Seattle vs Miami report

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DATA 5100

The Question

Seattle and Miami are two U.S. coastal cities with very different climates. Seattle is cool and marine, and Miami is warm and tropical. Both are known for their rain, but in different ways: Seattle for frequent drizzle, and Miami for heavy storms and a summer rainy season. This report compares rainfall in the two cities. Rainfall can be measured in different ways, so we look at two. The first is how much rain falls. The second is how often it rains.We also ask how the pattern changes through the year, and whether the two cities have rain in the same months or in different ones. Comparing them helps show what rain is actually like in each place, beyond their reputations.

Data

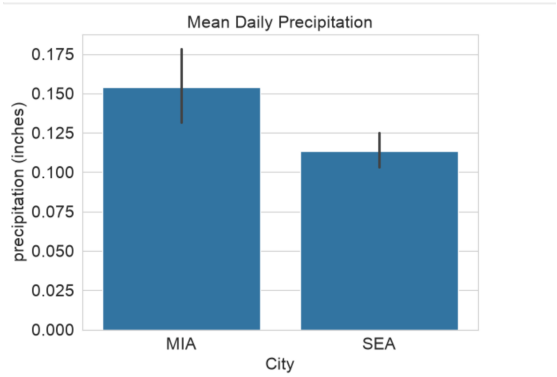
The data are daily rainfall measurements, in inches, from one weather station in Seattle,Washington and one in Miami ,Florida.Both cover 5 years from 1st of January 2018 to 31st of December (about 1800 days per City).A day with no rain is recorded as 0.0 .
There are days when there is no recorded measurement (175 in Seattle and 215 in Miami).
The monthly comparison need a value for every day,I filled the gap with the average rainfall recorded on the same day in the other years.Each city is represented by a single station, so the results describe those locations rather than every part of either city.

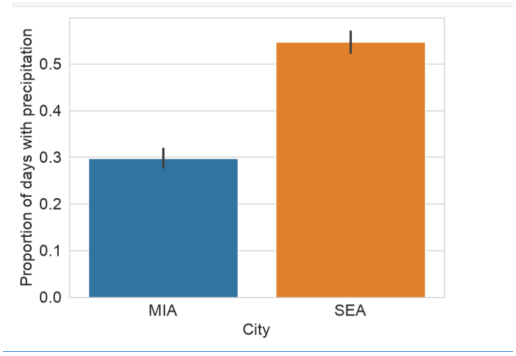
Methods

I first plotted the daily data to check what columns were reasonable to use and finding missing values.I then summarized rainfall in two ways:(1) **Mean daily precipitation for each city** and (2) **proportion of days with any rain.** I calculated both for the whole five years and for each calendar month, to capture seasons.Finally, for each month I see the gap between the cities was large enough that it is unlikely to be explained by chance alone, using statistics methods like t-test for rain amounts and z-test for proportion of days . The difference is “significant” when chance would produce a gap of more than 5%.Significant months are marked with a * in the graphs

Results

(1) **Generally it rains more days in Seattle but Miami averages more rainfall,** It rains about 50% of the days in Seattle and around 30% in Miami. In Miami we get 0.15 inches on average rainfall meanwhile in Seattle its around 0.115 inches of average rainfall.So we could conclude that Miami rain come in heavier bursts while Seattle rain tends to be light and frequent.

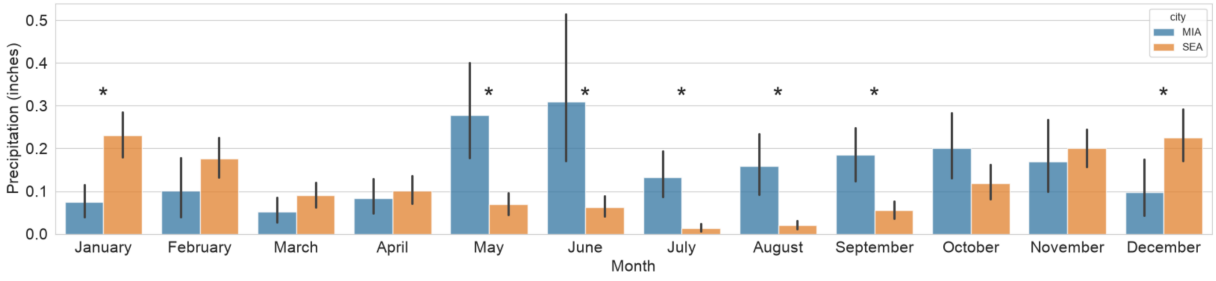




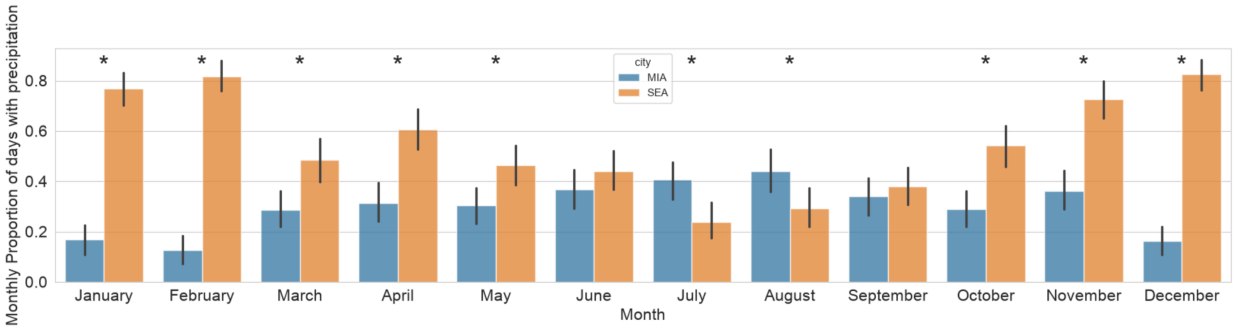
Fig(1): Mean daily Precipitation

Fig(2) Proportion of days with precipitation

(2)**Seattle has more rain and more number of days in the winter, spring season**,In months of November ,December, January ,February,March and April.Except in July and August ,every other month Seattle has more days with precipitation.The ones with significant difference is marked with a asterisk(*).



Fig(3) Precipitation for each month



Fig(4) Monthly proportion of days with precipitation

Conclusion

Seattle's reputation holds in one sense: it rains on far more days there, especially from Novembor through April, so a visitor in winter should expect rain on most days. But Seattle's rain is light. Miami receives more total rain, mostly in short, heavy downpours during its summer rainy season. In short, Seattle is the rainier city by number of rainy days, and Miami is the rainier city by amount of rain.